

MEDICAL DIAGNOSIS

AIM

To write a Prolog program for medical diagnosis using symptoms and rules to identify a possible disease.

ALGORITHM

1. Start.
2. Define diseases and their symptoms.
3. Create rules connecting symptoms with diseases.
4. Query the symptoms or disease.
5. Match the available symptoms with the rules.
6. Display the possible diagnosis.
7. Stop.

SOURCE CODE

main :-

```
write('What is the patient's name? '),  
read_line_to_string(user_input, Patient),  
diagnose(Patient).
```

diagnose(Patient) :-

```
hypothesis(Patient, Disease),  
format('~w probably has ~w.~n', [Patient, Disease]),  
!.
```

diagnose(_) :-

```
write('Sorry, I don''t seem to be able to diagnose the disease. '), nl.
```

symptom(Patient, fever) :-

```
format('Does ~w have a fever (y/n)? ', [Patient]),
```

response(y).

symptom(Patient, rash) :-

**format('Does ~w have a rash (y/n)? ', [Patient]),
response(y).**

symptom(Patient, headache) :-

**format('Does ~w have a headache (y/n)? ', [Patient]),
response(y).**

symptom(Patient, runny_nose) :-

**format('Does ~w have a runny nose (y/n)? ', [Patient]),
response(y).**

symptom(Patient, conjunctivitis) :-

**format('Does ~w have conjunctivitis (y/n)? ', [Patient]),
response(y).**

symptom(Patient, cough) :-

**format('Does ~w have a cough (y/n)? ', [Patient]),
response(y).**

symptom(Patient, body_ache) :-

**format('Does ~w have body ache (y/n)? ', [Patient]),
response(y).**

symptom(Patient, chills) :-

```
format('Does ~w have chills (y/n)? ', [Patient]),  
response(y).
```

symptom(Patient, sore_throat) :-

```
format('Does ~w have sore throat (y/n)? ', [Patient]),  
response(y).
```

symptom(Patient, sneezing) :-

```
format('Does ~w have sneezing (y/n)? ', [Patient]),  
response(y).
```

symptom(Patient, swollen_glands) :-

```
format('Does ~w have swollen glands (y/n)? ', [Patient]),  
response(y).
```

hypothesis(Patient, measles) :-

```
symptom(Patient, fever),  
symptom(Patient, cough),  
symptom(Patient, conjunctivitis),  
symptom(Patient, runny_nose),  
symptom(Patient, rash).
```

hypothesis(Patient, german_measles) :-

```
symptom(Patient, fever),  
symptom(Patient, headache),
```

**symptom(Patient, runny_nose),
symptom(Patient, rash).**

hypothesis(Patient, flu) :-

**symptom(Patient, fever),
symptom(Patient, headache),
symptom(Patient, body_ache),
symptom(Patient, conjunctivitis),
symptom(Patient, chills),
symptom(Patient, sore_throat),
symptom(Patient, runny_nose),
symptom(Patient, cough).**

hypothesis(Patient, common_cold) :-

**symptom(Patient, headache),
symptom(Patient, sneezing),
symptom(Patient, sore_throat),
symptom(Patient, runny_nose),
symptom(Patient, chills).**

hypothesis(Patient, mumps) :-

**symptom(Patient, fever),
symptom(Patient, swollen_glands).**

hypothesis(Patient, chicken_pox) :-

**symptom(Patient, fever),
symptom(Patient, chills),**

```
symptom(Patient, body_ache),
symptom(Patient, rash).
```

```
% -----
```

```
% Read response
```

```
% -----
```

```
response(Reply) :-
```

```
    read_line_to_string(user_input, Input),
```

```
    string_lower(Input, Lower),
```

```
    string_chars(Lower, [First|_]),
```

```
    Reply = First.OUTPUT
```

```
:-
% c:/Users/Abinaya Sri J/OneDrive/Desktop/Prolog exercise/MEDICA
What is the patient's name? Abi
Does Abi have a fever (y/n)? y
Does Abi have a cough (y/n)? y
Does Abi have conjunctivitis (y/n)?
Does Abi have a fever (y/n)? y
Does Abi have a headache (y/n)? y
Does Abi have a runny nose (y/n)? y
Does Abi have a rash (y/n)? y
Abi probably has german_measles.
```

RESULT

The Prolog program successfully demonstrates a simple rule-based medical diagnosis system by matching symptoms with predefined disease rules.